

## D&amp;P: LONGEST COMMON SUBSEQUENCE

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**Problem:** Given two sequences  $X = \langle x_1, x_2, \dots, x_m \rangle$  and  $Y = \langle y_1, y_2, \dots, y_n \rangle$ , find a common subsequence of  $X$  and  $Y$  that has the maximum length. The sequences can have any alphabet, symbol, or number as long as characters are in order. For example, if  $X = \langle A, B, C, B, D, A, B \rangle$  and  $Y = \langle B, D, C, A, B, A \rangle$ , the sequence  $\langle B, C, A \rangle$  is a common subsequence of both  $X$  and  $Y$ . The subsequence  $\langle B, C, A \rangle$  is not a longest common subsequence (LCS) of  $X$  and  $Y$ ; however, since it has length 3 and the sequence  $\langle B, C, B, A \rangle$ , which is also common to both sequences  $X$  and  $Y$ , has length 4.

**Here are guided steps to help you while thinking about a solution:**

- (i) Identify the “last” question you need to answer in developing a solution.

In this question, we ask what is the last decision we make? For given sequences,  $X = \langle x_1, x_2, \dots, x_m \rangle$  and  $Y = \langle y_1, y_2, \dots, y_n \rangle$ , is  $x_m$  or  $(y_n)$  part of longest common subsequence?

- (ii) Define subproblems, express the solution to the overall problem in terms of the subproblems.

**- Subproblems:**  $c[i, j]$  the longest subsequence that can be obtained for the sequences  $X_i = \langle x_1, x_2, \dots, x_i \rangle$  and  $Y_j = \langle y_1, y_2, \dots, y_j \rangle$

**-Main problem:**  $c[m, n]$ : When  $i = m$  and  $j = n$  then we will find the optimal solution which is the longest common subsequence that can be obtained from the sequences  $X_m = \langle x_1, x_2, \dots, x_m \rangle$  and  $Y_n = \langle y_1, y_2, \dots, y_n \rangle$

- (iii) Define and prove optimal substructure.

Let  $X = \langle x_1, x_2, \dots, x_m \rangle$  and  $Y = \langle y_1, y_2, \dots, y_n \rangle$  be sequences, and let  $Z = \langle z_1, z_2, \dots, z_k \rangle$  be any LCS of  $X$  and  $Y$ .

- (1) If  $x_m = y_n$ , then  $z_k = x_m = y_n$  and  $Z_{k-1}$  is an LCS of  $X_{m-1}$  and  $Y_{n-1}$ .
- (2) If  $x_m \neq y_n$  and  $z_k \neq x_m$ , then  $Z$  is an LCS of  $X_{m-1}$  and  $Y_n$ .
- (3) If  $x_m \neq y_n$  and  $z_k \neq y_n$ , then  $Z$  is an LCS of  $X_m$  and  $Y_{n-1}$ .

**Proof (1):** If  $z_k \neq x_m$ , then we could append  $x_m = y_n$  to  $Z$  to obtain a common subsequence of  $X$  and  $Y$  of length  $k + 1$ , contradicting the supposition that  $Z$  is a longest common subsequence of  $X$  and  $Y$ . Thus, we must have  $z_k = x_m = y_n$ . Now, the prefix  $Z_{k-1}$  is a length  $k - 1$  common subsequence of  $X_{m-1}$  and  $Y_{n-1}$ . We wish to show that it is an LCS. Suppose for the purpose of contradiction that there exists a common subsequence  $W$  of  $X_{m-1}$  and  $Y_{n-1}$  with length greater than  $k - 1$ . Then, appending  $x_m$  and  $y_n$  to  $W$  produces a common subsequence of  $X$  and  $Y$  whose length is greater than  $k$ , which is a contradiction.

**Proof (2):** If  $z_k \neq x_m$ , then  $Z$  is a common subsequence of  $X_{m-1}$  and  $Y$ . If there were a common subsequence  $W$  of  $X_{m-1}$  and  $Y$  with length greater than  $k$ , then  $W$  would also be a common subsequence of  $X_m$  and  $Y_n$ , contradicting the assumption that  $Z$  is an LCS of  $X$  and  $Y$ .

**Proof (3):** The proof is symmetric to (2).

- (iv) Formulate a recursive solution to the subproblems. Do not forget to specify the base case(s).

For  $1 \leq i \leq m$  and  $1 \leq j \leq n$ :

$$c[i, j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ c[i - 1, j - 1] + 1 & \text{if } i, j > 0 \text{ and } x_i = y_j \\ \max\{c[i, j - 1], c[i - 1, j]\} & \text{if } i, j > 0 \text{ and } x_i \neq y_j \end{cases}$$

- (v) Provide the pseudo-code of the bottom-up procedure you use to compute the value of the optimal solution, as well as the procedure for reconstructing the optimal solution.

Now, we have a matrix  $b$  which contains all of the LCS for each subproblem. In order to print the LCS, we can use the following function:

- (vi) Characterize the runtime of the resulting procedure assuming that you would implement your solution using a **bottom-up procedure**.

We need to consider

1. The number of subproblems
2. The amount of work done at each subproblem

There are  $m$  subproblems and  $n$  options for each subproblem. Therefore, the run time is:

$$T(n) = \Theta(nm)$$

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**Algorithm 1** LCS-LENGTH( $X, Y, m, n$ )

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1:  $b, c \leftarrow$  matrix of size  $m \times n$ 
2: for  $i \leftarrow 1$  to  $m$  do                                 $\triangleright$  First Base Case: second sequence it empty
3:    $c[i, 0] \leftarrow 0$ 
4: end for
5:
6: for  $i \leftarrow 1$  to  $n$  do                                 $\triangleright$  First Base Case: first sequence it empty
7:    $c[0, j] \leftarrow 0$ 
8: end for
9:
10: for  $i \leftarrow 1$  to  $m$  do                                 $\triangleright$  compute table entries in row-major order
11:   for  $j \leftarrow 1$  to  $n$  do
12:     if  $x_i = y_j$  then
13:        $c[i, j] = c[i - 1, j - 1] + 1$ 
14:        $b[i, j] = "$   $\nwarrow$   $"$ 
15:     else if  $c[i - 1, j] \geq c[i, j - 1]$  then
16:        $c[i, j] = c[i - 1, j]$ 
17:        $b[i, j] = "$   $\uparrow$   $"$ 
18:     else
19:        $c[i, j] = c[i, j - 1]$ 
20:        $b[i, j] = "$   $\leftarrow$   $"$ 
21:     end if
22:   end for
23: end for
24:
25: return  $c$  and  $b$ 
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**Algorithm 2** PRINT-LCS( $b, X, i, j$ )

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1: if  $i == 0$  or  $j == 0$  then
2:   return                                                 $\triangleright$  the LCS has length 0
3: end if
4: if  $b[i, j] == "$   $\nwarrow$   $"$  then
5:   PRINT-LCS( $b, X, i - 1, j - 1$ )
6:   print  $x_i$ 
7: else if  $b[i, j] == "$   $\uparrow$   $"$  then
8:   PRINT-LCS( $b, X, i - 1, j$ )
9: else
10:  PRINT-LCS( $b, X, i, j - 1$ )
11: end if
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